

Project Demo Planning

Date	15 March 2026
Team ID	TM-OPTICROP-04
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Project Demo Planning:

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities of each team member.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction & Slides	Explain current problems in traditional crop management and present the architectural overview of the OptiCrop engine.	2	Team Lead
2	Telemetry Ingestion	Showcase live data streaming from IoT simulated sensors into the real-time Soil Dashboard panel.	2	IoT Backend Dev
3	ML Yield & Scheduler	Run live user query scenarios through the XGBoost predictive models to display optimized yield and irrigation schedules.	3	ML Engineer
4	Leaf Disease Vision Hub	Upload a plant crop image to test the live inference response speed and validation scores of the ResNet50 framework.	2	CV Specialist
5	Alerts & Optimization Prescriptions	Demonstrate dynamic SMS dispatch triggers for weather anomalies and output exact fertilizer formulation recipes.	2	Backend Developer
6	Conclusion & QA Setup	Summarize future expansion modules, highlight overall system scalability, and open the floor for evaluator review questions.	4	All Members

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	

Step	Activity	Notes
		Emphasize agricultural yield losses due to unoptimized irrigation and incorrect soil chemical compositions. Keep slide transition crisp.
2	Solution Overview	Walk through the modular tech stack, highlighting how the telemetry database interacts cleanly with our predictive ML engines.
3	Live Feature Demonstration	Run comprehensive application paths including real-time sensor panel monitoring, fertilizer recommendations, and leaf image diagnostic upload.
4	Q&A Session	Prepare technical justifications for choice of XGBoost/ ResNet50 models, API response times, and target edge optimization strategies.